Oriented Ship Detection Based on Strong Scattering Points Network in Large-Scale SAR Images

Yuanrui Sun, Xian Sun, *Senior Member, IEEE*, Zhirui Wang, and Kun Fu

***Abstract*— Ship detection has broad applications in many areas, including fishery management, maritime rescue, and maritime monitoring. Recently, numerous detectors based on deep learning have been carried in ship detection in synthetic aperture radar (SAR) images. However, detecting the inshore ships faces enormous challenges because of the strong scattering interference of the inland area. In order to address such issues, a novel method named strong scattering points network for ship detection is proposed in this article. First, according to the SAR imaging mechanism, the ships usually appear strong scattering phenomenon in the SAR images. Therefore, the proposed method detects the strong scattering points on the ship and then aggregates their positions to obtain the ship's arbitrary orientation box. Second, our method designs an embedding vector to cluster these points as an individual object to regress the oriented bounding box. Third, in order to distinguish the strong scattering points on land, a ship attention module is employed to extract the image texture features and representations of local features. It can suppress the false alarm caused by land interference in the detection process. Furthermore, to demonstrate the effectiveness of the proposed algorithm, this article introduces a new ship dataset for oriented ship detection named large-scale dataset for ship detection in SAR images (LDSD). Moreover, the public SAR ship detection dataset (SSDD) is utilized to verify the robustness and generalization ability of the detector. The experimental results on two datasets show that our method has a strong anti-interference ability in the inshore background and achieves state-of-the-art detection performance.**

***Index Terms*— Deep learning, imaging mechanism, ship detection, strong scattering, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is an all-time and all-weather remote imaging sensor and has broad application prospects in target discrimination [1], [2], object detection [3], [4], ocean exploitation [5], terrain classification [6], and disaster prevention [7]. Ship detection is an important branch of object detection with great significance, which can monitor the water transportation in coastal areas, harbor management, and guarantee maritime safety [8]. As more and more satellites have been launched, such as Gaofen-3, TerraSAR-X, and Sentinel-1, a wide range of SAR images can be acquired and lay the foundation for the SAR scientific research and applications.

With the exploration of deep learning in many aspects by many researchers, the convolutional neural network (CNN) has been widely used in natural scene object detection and gained significant improvement. In comparison to the traditional ship detection method represented by the constant false alarm rate (CFAR), CNN-based methods can generate more feasible results because of the powerful representation ability and automatic feature extraction [9] in complex background conditions. However, the development of deep learning in ship detection is restricted owing to the enormous domain differences between natural and SAR images [10]. While these natural scene-based detectors have been proved effective in detecting SAR ships [11] to some extent, there still exist several limitations in object detection. First, the physical significance of pixel value in SAR images is different from that in optical images. Optical images represent the color information, while SAR images are electromagnetic scattering information. The main CNN-based detectors [12]–[16] are mainly designed for natural scene objects. Natural scene image has abundant color information, while SAR image pixels represent the reflected electromagnetic wave intensity of the object. Therefore, these detectors without the combination of SAR image characteristics usually result in high false alarm and weak robustness. And it needs a deep learning-based detection algorithm designed for SAR image features. Second, object detection in natural scene has numerous categories, while ship detection mainly focuses on one category. Therefore, the huge diversity in the number of categories leads to the above detectors cannot be optimized for the "ship" category. Third, the deep learning detectors based on anchors have many parameters to adjust, and these anchor parameters are designed for natural scene objects, such as cats, dogs, chairs, and so on. It is improper to employ these anchor parameters directly in the SAR ship detection task.

In recent years, various methods of SAR ship detection have been proposed [17]–[20] and remarkable progress has been made. However, ship detection has two major problems. First, only a few methods are proposed for oriented ship

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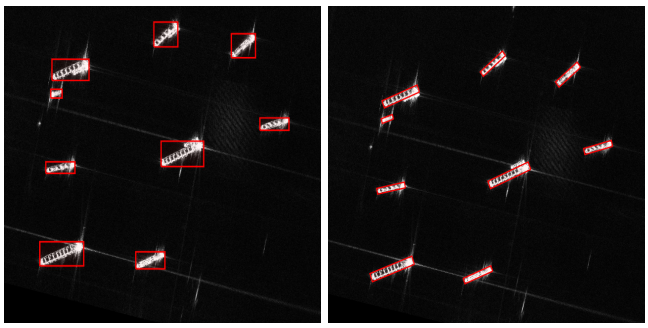


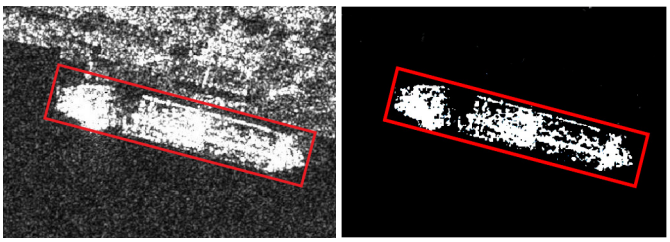
(a) (b)

Fig. 1. (a) Horizontal and (b) oriented bounding box of ships.

detection. The traditional ship detection methods and deep learning-based approaches utilize the horizontal bounding box, which cannot fit the narrow and long ship well and is difficult to reflect the direction information of the ship. As the ship may be docked at any angle, the horizontal detection box inevitably contains some redundant backgrounds, which is displayed in Fig. 1(a). In contrast, the oriented bounding boxes in Fig. 1(b) can pay more attention on the ship instead of the background. Second, inshore ship detection is a challenging task. Because the complex land area, such as harbor facilities and buildings, usually appears strong scattering phenomenon, which is similar to the ship imaging result, the strong interference on land generally results in lots of false alarms.

In order to solve these problems, many deep learning-based methods have been proposed in SAR ship detection [21]–[24]. Although these approaches have been proved to be effective, there still exist several shortages. First, the existing methods are mainly divided into two parts. One is based on pre-defined anchor box methods, whereas the other is anchor-free methods. To reach high detection performance, the former usually samples the input image by densely scales and aspect ratios anchor boxes. This produces a heavy amount of hyperparameters and computation. Moreover, the detection recall is degraded because the predefined anchor box is not suitable for the new detection scene. The latter is usually unable to effectively extract the features of the ship in the SAR image due to the lack of combining SAR image features, which results in low localization accuracy. Second, the physical meaning of the SAR image is different from natural images. The same object may have different characteristics in an image under different environmental conditions. These detectors are not robust to the above changes. Therefore, it is necessary to delve into the SAR image features to handle this phenomenon.

Another point of our work is SAR ship detection in large-scale images. There are two main challenges in this field. First, in large-scale SAR images, there are more interferences from lands or islands, and the background around the ship is complex. Some irrelevant objects, such as inland facilities, are likely to be misidentified as the ships because of similar geometry, which results in a quantity false alarm. Therefore, building an end-to-end and practical model is a challenging task. Second, the average distribution of ship targets in large-scale SAR images is sparse. Only a few regions are positive samples (containing ships) while most positions in the image are negative samples (background). The imbalance of positive and negative samples puts forward a high requirement for the ability of the feature extractor.



(a) (b)

Fig. 2. (a) Original SAR ship image. (b) Ship strong scattering points.

For tackling the above issues, a ship detection method utilizing SAR characteristics is proposed in this article. As shown in Fig. 2(a), most ships are made of metal shells that have strong scattering of electromagnetic pulses and exist a large number of strong scattering points in SAR image. As shown in Fig. 2(b), when these strong scattering points are extracted, there are ample key points on the ship to calculate the corresponding minimum bounding rectangle (MBR). This article is inspired by this phenomenon and proposes a strong scattering points network for oriented ship detection in combination with the ship imaging characteristics. First, the proposed method utilizes the ResNet as the backbone to extract image features, which are widely used in deep learning by powerful representation ability. Second, to enhance the detection capability of multi-scale ships, this article designs the architecture atrous convolution stack (ACS) module. This module stacks different dilate rates of convolution to learn multi-scale information about ships. As a consequence, the detection performance is increased by improving the adaptability to different scale ships. Third, this article employs a ship attention module to extract the texture distribution of strong scattering points and try to distinguish the difference between the ship target and the land area. This module enhances the robustness of model and suppresses the interference on land. Then, in order to obtain the bounding box, the strong scattering points are detected and the designed grouping network can gather these points in feature maps by generating embedding vectors. Furthermore, this module is based on an anchor-free mechanism, which decreases the number of network hyperparameters.

To assess the effectiveness and conduct in-depth capacity analysis of the proposed method, we build a dataset based on Chinese Gaofen-3 satellite named large-scale dataset for ship detection in SAR images (LDSD). Comprehensive evaluations on the LDSD and the SAR ship detection dataset (SSDD) [25] prove that the proposed method has the ability to reach the highest localization accuracy and possess the strong robustness capability for the ship detection task.

The major contributions to our work are summarized as below.

1) In this article, a novel method of detecting ships in SAR images of large scale is proposed. According to the SAR imaging mechanism, this method exploits the ship

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characteristics and extracts the strong scattering points for ship detection in SAR images. After obtaining strong scattering points, the oriented bounding box is regressed to achieve more accurate object detection results. And in LDSD, the proposed method attains state-of-the-art performance.

2) In order to reduce false alarm in the inshore strong interference environment, this article proposes the ship attention module. Due to the attention module, the discrimination between ship and land is increased and anti-interference of our detection algorithm is improved.
3) The proposed ACS module stacks the atrous convolution to extract abundant multi-scale features and context information. In this procedure, sufficient multi-scale information is extracted to increase the localization ability for the ships of various sizes.

The rest of this article is organized along the following lines. Section II briefly reviews the related works. Section III describes the details of the proposed method. Section IV introduces the LDSD dataset used in the experiments and the analysis of ship detection results on the LDSD. Finally, the conclusion is drawn in Section V.

II. Related Works

This section summarizes the previous works in the SAR ship detection field. The brief descriptions of some traditional ship detection methods, deep learning-based models for object detection in the natural scenes, and SAR images are provided below, respectively.

A. Traditional Ship Detection Methods

For increasing detection rate and reducing the false alarm, the remote-sensing images are usually preprocessed before using the ship detection algorithm. At the same time, the detection results need to be post-processed. Generally, ship detection is divided into four steps.

1) *Land masking:* The purpose is to cover the land area so that the ship detector only works in the ocean area.
2) *Preprocessing:* The purpose is to remove the clutter background, highlight the ship area, eliminate the interference of various factors on ship detection, and increase the accuracy and dependability of ship detection as much as possible.
3) *Object detection:* Using the ship detection approaches to detect the suspected object in the ocean area.
4) *False alarm suppression:* Its primary purpose is to screen the suspected objects obtained by detection through prior knowledge and manual intervention.

As for the detection methods mentioned in Step 3, before the popularity of deep learning, researchers from various countries conducted many kinds of studies in the area of SAR ship detection and presented numerous classical methods [26]–[32]. The classical algorithms focus on the artificial design of feature extractors to generate ship features, which generally have lower performance and less discriminative compared with deep learning-based algorithms [33], [34]. Novak *et al*. [26] propose a dual-parameter CFAR method using the Gaussian model method. Stagliano *et al*. [27] also develop an algorithm based on CFAR and wavelet transform. He *et al*. [28] further present a polarization SAR ship automatic detection method, which generates multi-scale super-pixels to calculate the measured value between a super-pixel and surrounding pixels. Kapur *et al.* [31] use the Shannon entropy concept for image segmentation. The main idea is to maximize the informativeness between the object and the background in the image. This method overcomes the single-parameter threshold segmentation algorithm's problems in detecting unconnected ships. As for CFAR, the core idea of this algorithm is to keep the false alarm rate constant. According to the false alarm rate and the statistical characteristics of sea clutter, the threshold of ship object detection is calculated.

When a Gaussian model is used to establish an ocean background clutter model, a two-parameter CFAR algorithm [29], [30] can be obtained. In many cases, the Gaussian model's description of ocean clutter is not ideal, so some researchers introduced K distribution [32] to build a model of ocean clutter. However, the above-mentioned detectors heavily rely on ocean clutter modeling. Due to the complexity of the practical application scene, it is a challenging task to model the inshore scene. Therefore, they often perform better in detecting offshore ships with a simple backgrounds, but the results in the inshore ships with complex backgrounds are not satisfactory.

B. Deep-Network-Based Object Detectors

Recently, owing to the contribution of many researchers to deep learning, a variety of methods have been presented for object detection, which is mainly separated into two categories: one-stage object detector and two-stage object detector.

Representative algorithms of one-stage object detectors include SSD [12], YOLOv1 [13], RetinaNet [14], CornerNet [35], and FCOS [36]. They combine the extraction and detection and directly get the result of object detection. YOLOv1 separates the original image into $S \times S$ grid. Each object's grid center is in charge of predicting the position of the object and category. However, one grid center can only predict one class of objects. The improvement of SSD lies in the introduction of anchor module and multi-scale feature extraction layer, which addresses the shortcomings of the YOLOv1 rugged grid and low detection precision for small objects. RetinaNet points out that the major reason for the poor performance of one-stage detector is the imbalance in the selection of positive and negative samples. Therefore, the focal loss is proposed to significantly improve the accuracy of the one-stage detection algorithm. CornerNet considers object detection task as an issue of detecting key points. From the CornerNet point of view, the predicted bounding box is obtained by detecting corner points of the object. FCOS mainly performs pixel-by-pixel prediction and removes the pre-defined anchor box to avoid complex calculations related to anchor boxes, which significantly reduces the amount of parameters and memory.

Representative two-stage methods consist of Faster-RCNN [15], feature pyramid networks (FPNs) [37], R-FCN [38], Mask-RCNN [16], and Cascade-RCNN [39]. A series of

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proposal boxes are generated for the candidate region using the above methods, then categorize the proposal boxes, and further regress the accurate position of the proposal boxes. The fundamental difference with one-stage methods is that the two-stage methods have one more candidate boxes extraction process. The appearance of Faster-RCNN is an important milestone of the two-stage detectors. It is formed by three parts: the backbone network, the region proposal network (RPN), and the bounding box regression network. Benefitting by the RPN, the detection accuracy is significantly improved. By contrast, it also dramatically raises the cost of detection time. After Faster R-CNN, Mask-RCNN is proposed to improve the performance of object detection. The main contributions of Mask-RCNN are to introduce FCN to generate the relevant mask branch and propose a corresponding RoIAlign solution for the pixel deviation problem in RoI Pooling. Cascade-RCNN uses a multistage structure, and each stage uses a different intersection over union (IoU) threshold for training, which has a good optimization effect on RPN proposals and achieves state-of-the-art result in the COCO dataset. In order to capture the feature of the small-scale objects and achieve satisfactory performance, many researchers present different multi-scale feature extraction modules. FPN [37] constructs a rich, multi-scale feature pyramid from a single-resolution input image and each level of the pyramid can be used for detecting objects at a different scale. The ASPP [40] employs different rate atrous convolutions to extract multi-scale features. TridentNet [41] mainly solves the problem of scale variation for object detection. It designs a multi-branch structure and scale-aware training method to generate scale-specific feature maps. Libra R-CNN [42] presents a novel balanced feature pyramid to possess balanced information from each resolution. However, they concentrate primarily on natural optical images. It is inappropriate to use the algorithms mentioned above in SAR images owing to the enormous domain mismatch. A better solution for addressing this critical issue is to find out the difference between SAR images and natural optical images and use the characteristics of SAR image to detect ship objects.

C. Object Detection in SAR Images

Most deep learning-based models, as mentioned above, are designed for natural scene object detection. Using deep CNNs-based algorithms in SAR ship detection is still a challenging task. Moreover, many researchers have been proposed deep CNNs-based algorithms for object detection in remote sensing images [43]–[49]. And many of them are orientation detectors. Cheng *et al*. [43] introduce a new rotation-invariant layer on CNN architecture for geospatial object detection. Ding *et al*. [44] propose a learnable rotated RoI module which can transform horizontal RoI to rotating RoI. Fu *et al.* [49] present a coarse-to-fine fashion orientation estimator for object detection in remote-sensing image. Although the SAR image and optical remote-sensing image have similar shooting angle, terrain, and target, they have a quite difference in data distribution. Directly using these methods in SAR ship detection will get poor results.

Recently, various CNNs-based algorithms have been proposed to address these problems. Kang *et al*. [21] fuse high-resolution RPN and contextual features of SAR ship to promote the localization accuracy for small-scale ships. Jiao *et al*. [22] proposed a new SAR ship detection algorithm using densely connected multi-scale neural network. Chen *et al*. [23] introduce two parallel attention module branches and conduct an effective loss function to increase the generalization of the network. In order to suppress the false alarm and capture the ship features, some studies adopt the attention mechanism. Cui *et al*. [11] propose a dense attention pyramid network and concatenate feature map from top to bottom of the pyramid to detect multi-scale SAR ship. Fu *et al*. [50] introduce an attention guided balanced pyramid to semantically balance the features across different pyramidal levels. Cui *et al*. [51] present a spatial shuffle-group enhance attention module to get semantic features. Du *et al*. [4] draw saliency information of SAR images into SSD detector to suppress clutter and highlight interesting objects. Zhao *et al*. [24] fuse multibranch dilated convolution and convolution block attention to discrete scattering features and highlight significant features of aircrafts. However, these methods lack a strategy to fuse adequate characteristics of SAR image and ship detection task. Therefore, a method using strong scattering points is involved in this article.

III. Methodology

A description of the proposed method will be introduced in this section. Fig. 3 illustrates the detailed architecture of the proposed method. The proposed method has four components: the backbone network, the ACS module for extracting multi-scale ship features, the ship attention module to suppress false alarms, and the detection and grouping module for final prediction. In this section, this article will introduce each module in detail.

A. Atrous Convolution Stack Module

Deep convolutional neural networks (DCNNs) [52] composed of fully convolution [53] has been proved to be adequate in object detection task. However, the application of max-pooling in the convolution layer and the repeated striding operation vastly decrease the spatial information of the feature map [54]. Instead, in this article, we employ "atrous convolution" to release this problem. Additionally, multi-size SAR ship detection in complex scenes requires high localization ability of the algorithm. Therefore, atrous convolution is stacked to extract sufficient multi-scale features to increase the localization ability of the algorithm in this article. Next, we will introduce the ACS module in detail.

We utilize ResNet101 [55] as the backbone to extract image features. As known, the output stride of the feature map is 16, which means the feature map's resolution is reduced by 16 times across each direction compared with the input image. Some researchers have used deconvolutional layers to recover the spatial resolution [56]–[58], but this way adds extra memory and cost time. Instead, atrous convolution has been proved more effective in DCNNs as a potent tool in

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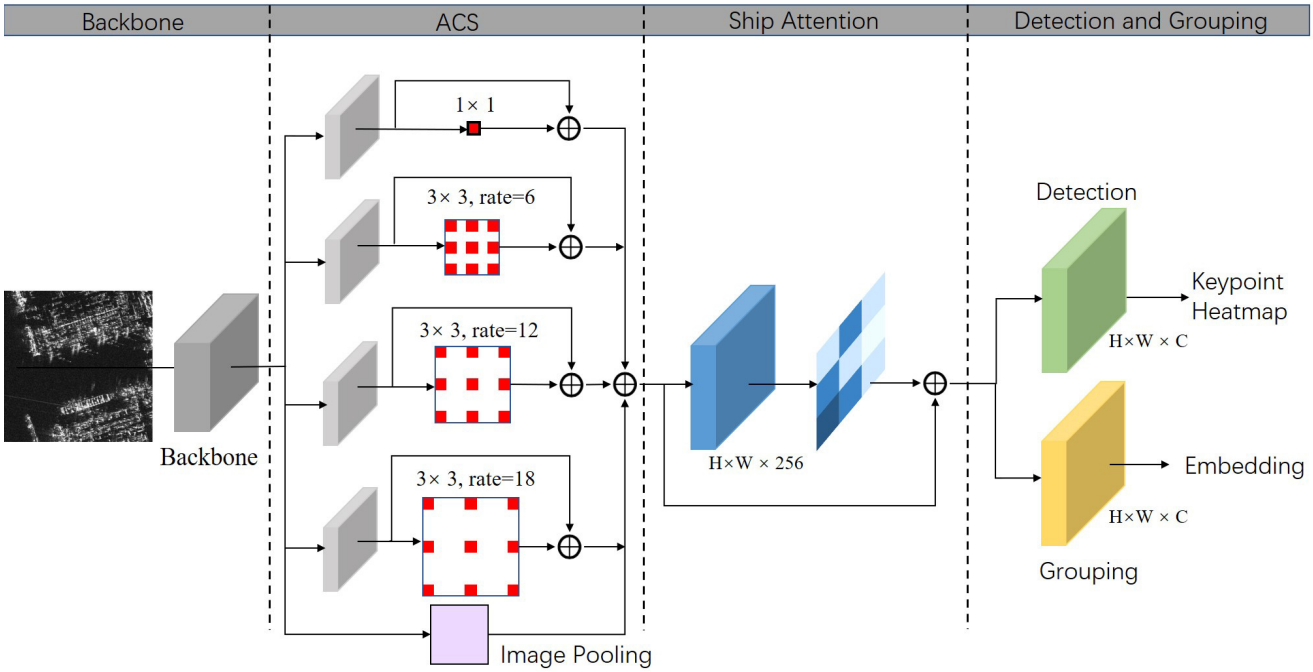

Fig. 3. Overall architecture of our method in this article.

dense ship detection tasks [40]. Meanwhile, the heatmap size is equal to the input image size. The strong scattering points in the shipping area are all regarded as key points to predict, while the number of the key points will be enough to form a dense prediction area. Therefore, the proposed method chooses atrous convolution to expressly rule the resolution of the feature map and significantly expands the receptive field of filters to accommodate a wider multi-scale context feature without adding additional parameters and computing time.

Consider a 2-D feature map, this article employs $K_c \times K_c$ atrous convolution over the input feature map $\boldsymbol{x}$ as

$$\boldsymbol{y}[\boldsymbol{i}][\boldsymbol{j}] = \sum_{\boldsymbol{m}=1}^{K_c} \sum_{\boldsymbol{n}=1}^{K_c} \boldsymbol{x}[\boldsymbol{i} + r\boldsymbol{m}][\boldsymbol{j} + r\boldsymbol{n}]\boldsymbol{w}[\boldsymbol{m}][\boldsymbol{n}] \tag{1}$$

where $\boldsymbol{i}$, $\boldsymbol{j}$, and $\boldsymbol{w}$ denote each location on the input feature map, the output, and convolution filter, respectively. And the atrous rate r matches the stride of the proposed method when sampling the input feature map, which can get more ship multi-scale context information and avoid the loss of resolution caused by striding operation by inserting $r-1$ zero between two consecutive filter values along the same spatial dimension. When $r = 1$, atrous convolution equals the regular convolution.

ACS is inspired by the structure of deeplab [54] and spatial pyramid pooling (SPP) [59], [60]. The experiments in SPP show that resampling convolutional features extracted at a single scale can significantly improve the performance of the network. ASPP utilizes multiple atrous convolution branches with different sampling rates to extract features in parallel and then fuses all branches to generate the final features. As shown in Fig. 3, the structure of ACS is similar to SPP, which is effective to capture multi-scale ship features. However, the study in article [54] shows that a too large rate value is inappropriate. In the worst case, it degrades the 3×3 filter to 1×1 filter, which reduces the number of valid filter weights and loses the internal correlation of the filter. To avoid using too large rate sampling and capture more global multi-scale ship context information, ACS is applied in our method. This module consists of one 1×1 convolution, three 3×3 convolutions with rates $= (6, 12, 18)$, and global average pooling. These parts are all applied to the feature map that output by the backbone. This module employs 1×1 convolution to increase the representational power of neural networks [61], [62]. Because of it, the ACS module can extract more multi-scale ship information. Different from the general ASPP [54], ACS applies shortcut connection in this module to help the network memorize the previous features and reduce the impact of negative samples. In the sea area, some amplitudes of the image are close to zero and the network outputs zero activation values (or extremely small activation value). After the iterative training, the gradient of the convolution layer decreases gradually and it is not conducive to the back propagation of other convolution layer gradients. The shortcut connection is equivalent to adding the constant one to the derivation and the transmission of other convolution layer gradients can be guaranteed [55]. Therefore, the ACS module can reduce the impact of negative samples on ship detection performance and improve the strong scattering point feature representation capability of the proposed model. As displayed in Fig. 3, each output of atrous convolution is added to the copied input features. The 1×1 convolution and 3×3 convolution in Fig. 3 are all 256 filters, stride one with batch normalization. After global average pooling, there is an 1×1

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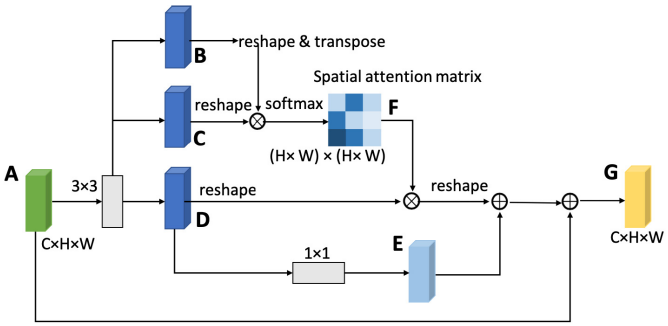

Fig. 4. Ship attention module.

convolution with 256 filters, batch normalization, and Relu layers. Results from all the atrous convolution are assembled by element-wise adding operation and then pass through a 3×3 filter.

B. Ship Attention Module

In SAR images, the strong scattering points not only exist in the ship, but also exist on inshore land. Therefore, this requires our algorithm to distinguish whether the strong scattering points belong to ships or inshore lands (buildings, harbors, and other metal objects). However, in optical images, many studies [63], [64] show that local feature representations generated by traditional DCNNs could misclassify objects and stuff. Thus, this article designs a ship attention module to highlight the shipping area and suppresses the false alarm in the land area to enhance the discrimination ability of the algorithm.

In SAR image, the ship is often surrounded by water and wharf, and the land area looks similar from the satellite perspective. Although the strong scattering points exist in many areas, the image texture of the shipping area is different from the land area. The artificial facilities in land usually have a highly repetitive and similar texture. To distinguish the texture between the ship and artificial facilities in land by local feature representations, this article introduces a ship attention module.

As illustrated in Fig. 4, after the ACS module, the output feature map with rich feature texture is got. When the input feature map $\mathbf{A} \in \mathbb{R}^{C\times H\times W}$ is obtained, the proposed method feeds it into 3×3 convolution layers to get three new feature maps $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$, where $\{\mathbf{B}, \mathbf{C}, \mathbf{D}\} \in \mathbb{R}^{C\times H\times W}$. The convolution layer consists of 3×3 filter, batch normalization, and Relu activation function. Then the proposed method reshapes feature maps $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ to $\mathbb{R}^{C\times M}$, where $M = H \times W$ is the size of the feature map. After that, the proposed method performs a matrix multiplication between the transpose of $\mathbf{C}$ and $\mathbf{B}$. Then, a softmax layer employs to compute the spatial attention feature map $\mathbf{F} \in \mathbb{R}^{M\times M}$

$$f_{ji} = \frac{\exp(B_i \cdot C_j)}{\sum_{i=1}^{M} \exp(B_i \cdot C_j)} \tag{2}$$

where f_{ji} denotes the influence exerted from ith position ($B_i \in \mathbb{R}^C$) to jth position ($C_j \in \mathbb{R}^C$). The multiplication can be regarded as taking the correlation matrix. It can suppress the false alarms with highly repetitive and similar texture, such as human-made facilities inland, as far as possible to retain the part of the image texture with significant differences. Note that the feature representation of the two locations reflects their correlation. And the feature representation of land and ship is different. Due to this characteristic, the ship attention module helps our method to highlight strong scattering in the shipping area.

Next, the proposed method feeds the feature map $\mathbf{D}$ into an 1×1 convolution layer with batch normalization to get feature map $\mathbf{E} \in \mathbb{R}^{C\times H\times W}$ and multiply $\mathbf{F}$ by a contribution control parameter α. The 1×1 convolution layer increases the nonlinearity [61] of the ship attention module and makes the network more aware of the image texture features. Then, an element-wise adding operation with feature maps $\mathbf{E}$ and $\mathbf{A}$ is employed to generate the result $\mathbf{G} \in \mathbb{R}^{C\times H\times W}$ as

$$G_{ij} = \alpha f_{ij} D_{ij} + \beta E_{ij} + A_{ij} \tag{3}$$

where α and β are initialized as 0 and automatically learn the optimal value. The learning strategy is the same as in [65]. Finally, the feature map $\mathbf{G}$ will go through the Relu layer to get the final output. From the above equation, the features of original location and all other locations are weighted sum in the final feature map $\mathbf{G}$. According to the spatial attention map, ship attention module selectively aggregates the context to highlight the strong scattering points in the shipping area and suppress the strong scattering points in the land area.

C. Detection and Grouping Module

After the ship attention module, it is the detection and grouping module, as shown in Fig. 5. This module is designed to obtain strong scattering points in SAR image and combine them into independent ship target. In this module, our method predicts the key points heatmap. This heatmap contains a series of key points of ship target and has one channel with the size of $H \times W \times 1$, where each position value of the heatmap represents the probability of this location become a ship key point. $H \times W$ represents the input image size. The probability value is continuously distributed from 0 to 1.

For each ship, the oriented bounding box is labeled by the expert as Fig. 1(a). In our method, the strong scattering points in SAR images are selected as following steps.

Step 1: Use the following formula to calculate the mean and variance of sea amplitude. N_s and x represent the number of pixels in the label area and the value of each pixel, respectively,

$$E(x) = \frac{1}{N_s} \sum_{i=1}^{N_s} x_i \tag{4}$$

$$\mathrm{Var}(x) = \frac{1}{N_s} \sum_{i=1}^{N_s} x_i^2 - (E(x))^2. \tag{5}$$

Step 2: Model the sea amplitude distribution. In [66], the K probability density function (pdf) has been found to be a particularly useful model for sea amplitude statistics. Therefore, this article employs K pdf to model the amplitude of high-resolution sea echo. Following previous work, the K pdf in SAR image is given by [67]:

$$f(x) = \frac{2}{x\Gamma(\upsilon)\Gamma(L)} \left(\sqrt{\frac{L\upsilon x}{\mu}}\right)^{\frac{L+\upsilon}{2}} K_{\upsilon-L}\left(2\sqrt{\frac{L\upsilon x}{\mu}}\right) \tag{6}$$

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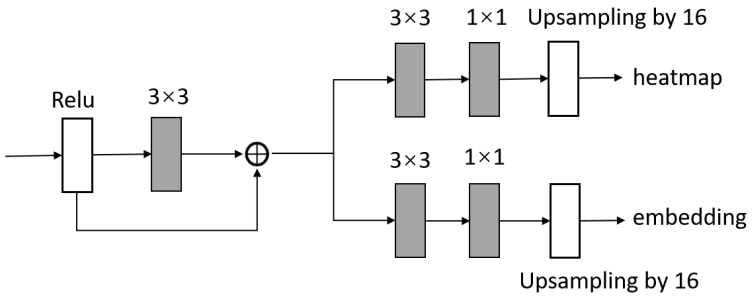

Fig. 5. Structure of detection and grouping module.

where μ is the image intensity mean, υ is a shape parameter, and L is the number of statistically independent looks. In this article, LDSD is a single-look SAR dataset. $\Gamma(\cdot)$ is the Eulerian gamma function, and $K_{\upsilon-L}(\cdot)$ is the modified Bessel function. And the relationship between $\mathrm{Var}(x)$, υ, and L can be expressed as follows [67]:

$$\mathrm{Var}(x) = \left[\left(1+\frac{1}{\upsilon}\right)\left(1+\frac{1}{L}\right)-1\right]\mu^2. \tag{7}$$

Step 3: Calculate the threshold T to extract strong scattering points. The proposed method calculates T by following formula [67]:

$$1 - P_{\mathrm{fa}} = \int_0^T f(x)dx \tag{8}$$

where P_{fa} represents the false alarm of extracting strong scattering points in the sea area. Then, according to [68], the cumulative distribution function and threshold T can be obtained. Each pixel value exceeding the threshold T in the labeled area is regarded as a strong scattering point. In this article, the value of P_{fa} is set to 0.1. It is worth noting that the strong scattering points are extracted only in the rectangular box region marked by ship. This operation can ensure that the false alarm in the process of extracting strong scattering points of ships is far less than P_{fa}. The process of strong scattering points selection is automatic and adaptive. The selected strong scattering points are shown in Fig. 2(b). After selecting strong scattering points, these key points are positive locations, and all other locations are negative. The detailed composition of the detection and grouping module is shown in Fig. 5. We first feed the input feature map to a residual block. This residual block contains 3×3 convolution layers with 256 channels, batch normalization, and Relu layers and performs an element-wise adding operation with the input features. Following the residual block, this module has two branches for predicting the heatmap and embedding vector. The residual block is fed into 3×3 convolution layers with 256 channels, batch normalization, Relu layers, and 1×1 convolution layers with one channel to produce the heatmap and embedding vector. The dimensions of heatmap and embedding vector are $H \times W \times 1$, while the dimensions of former feature maps are $(H/16) \times (W/16) \times 256$. Therefore, this module employs 1×1 filter to change the dimension [61] of the feature map to $(H/16) \times (W/16) \times 1$ and applies bilinear interpolation up-sampling operation to get the result.

The training loss function in our method consists of three parts

$$L = L_{\mathrm{det}} + \epsilon_1 L_{\mathrm{pull}} + \epsilon_2 L_{\mathrm{push}} \tag{9}$$

where ϵ_1 and ϵ_2 are the contributions for the pull and push loss. The proposed method sets both ϵ_1 and ϵ_2 to 0.1. In the part of predicting heatmap, the L_{det} loss function designed is as follows:

$$L_{\mathrm{det}} = -\frac{1}{N}\sum_{i=1}^{H}\sum_{j=1}^{W}\begin{cases}\left(1-p_{ij}\right)^{\gamma}\log\left(p_{ij}\right) & \text{if } y_{ij}=g_{ij}\\ \left(p_{ij}\right)^{\gamma}\log\left(1-p_{ij}\right) & \text{otherwise}\end{cases} \tag{10}$$

where L_{det} denotes the localization loss and N is the number of key points in an image. Meanwhile, p_{ij}, y_{ij}, and g_{ij} represent the score, prediction result, and ground truth at location (i, j) in the predicted heatmap, respectively. γ is the hyperparameter which controls each point's contribution. Our method set γ to be 2 in all experiments.

Multiple ships may exist in one SAR image, and multiple key points belong to different ships may be detected. The proposed method needs to determine whether a series of key points are from the same oriented box or not. Law and Deng [35] and Newell *et al*. [69] method for the task of paired key points detection and multi-person pose estimation inspired us to propose this module. Newell detects all human joints as a series of key points. Law detects a pair of the corner points to form the horizontal bounding box. They all establish an embedding vector for each key point and group them according to the distance between the embedding vectors. Establishing an embedding vector of key points is also practical to the proposed method. The proposed method makes provision for an embedding vector for each of the key points detected. If a series of key points belong to the same ship object, the distance of the same ship between their embedding vectors should be small. The proposed method follows Newell *et al.* [69] and Law and Deng [35]. We also use one dimension embedding vectors. Similar to [69], the proposed method defines the "pull" loss to train the network to gather the key points belonging to the same ship and the "push" loss to divide the key points belonging to other ships. The loss function of embedding vectors as follows:

$$L_{\mathrm{pull}} = \frac{1}{N}\sum_{k=1}^{N}\frac{1}{N_k}\sum_{i}^{N_k}\left(e_{k_i}-e_k\right)^2 \tag{11}$$

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TABLE I
DATASET INFORMATION

Resolution	Imaging Mode	Polarization	Azimuth Pixel	Range Pixel	Number of Images
1m	SL	VV	12000	15000	4
1m	SL	VV	26000	26000	5
1m	SL	HH	25000	30000	60
1m	SL	HH	27000	32000	26
1m	SL	HH	28000	38000	9
1m	SL	HH	25000	25000	23

$$L_{\text{push}} = \frac{1}{N(N-1)} \sum_{k=1}^{N} \sum_{\substack{s=1 \\ s \neq k}}^{N} \frac{1}{N_s} \sum_{j}^{N_s} \max\left(0, \Delta - \left|e_k - e_{s_j}\right|\right) \tag{12}$$

where e_{k_i} denotes the embedding vector for the i key point belonging to the ship k, and e_k is the average embedding vector of all the key points belonging to the ship k, and e_{s_j} denotes the embedding vector of the key points that do not belong to the ship k. N, N_k, and N_s correspond to the number of ships as well as the number of scattering points of the ship k and s, respectively. Our method sets Δ to be 2 in all experiments.

In this module, we get the network's final output heatmap and embedding vector. According to the heatmap output, our method can get the ship key point locations and group them based on the distances of embedding vectors. First, our method chooses the key point with the highest confidence in heatmap as the starting point to group, and the embedding vector distances less than a threshold τ (we set to 0.5) of other key points are all grouped as the same ship. Then our method chooses the highest confidence key point in the rest locations and repeats the previous step until all the key points are grouped. Additionally, the real values in the embedding vectors are immaterial. This module groups the key points by calculating the distance between the embedding vectors. The MBR of a series of key points belonging to the same object is taken as the oriented bounding box of the ship. Finally, the MBR is processed by non-maximum-suppression (NMS) [70] to get the final oriented bounding box. This article sets the threshold of NMS to 0.3 that can help us to remove the overlap box.

IV. EXPERIMENTS

A. Dataset and Settings

To demonstrate the effectiveness of the proposed approach, we conduct the experiments on the dataset named LDSD. In addition, to further verify the robustness and generalization ability of the detector, SSDD [25] is utilized for comparison experiments. The datasets are described as below.

1) LDSD: LDSD is a large-scale dataset consisted of 126 single-polarization SAR images from the Chinese Gaofen-3 satellite with expert annotation. The detailed information of LDSD is listed in Table I. Several image samples in the dataset are shown in Fig. 6. The imaging resolution of LDSD is 1 m, and the imaging model is spotlight. The polarization includes VV and HH. Additionally, the image format is tiff and the image size ranges from 12 000 × 12 000 pixels to 40 000 × 40 000 pixels. There are 6381 inshore ships in various environmental conditions in LDSD. The dataset is divided into two parts at random: the training set and the testing set. The training set contains 98 SAR images, and the testing set contains 28 SAR images. In the experiments, our method and other detectors use the training set to train and compare the performance on the testing set.

2) SSDD: It consists of 1160 images and 2456 labeled ships in various scenes. The image data is collected from RadarSat-2, TerraSAR-X, and Sentinel-1. There are four polarization modes: HH, HV, VV, and VH. The image resolution ranges from 1 to 15 m. SSDD is divided into training, validation, and test set with the proportion of 7:1:2.

B. Implementation Details

The backbone network of the proposed method and other detectors is ResNet101 for all the experiments. The strong scattering points network finetunes the pretrained ResNet101 model weights in training set to adapt to the object detection task. After the backbone, other convolutional layers are utilized to the same initialization method, and the training strategy of the ship attention module is followed by Zhang *et al.* [65]. The strong scattering points network is trained using the stochastic gradient descent (SGD) algorithm [52]. The initial learning rate is set to 0.0002 for the first 80 epochs and decreases to 0.00002 for the next 40 epochs. The momentum and weight decay of SGD are set to 0.9 and 0.0001, respectively. The dataset LDSD is single look, and this article does not apply any noise suppressing filter to the images because the proposed method is robust enough. Considering the conflict between the large size of SAR images in LDSD and restricted GPU memory, it is improper to directly train the strong scattering points network on the raw large-scale SAR images. Therefore, this article crops the large-scale SAR images into more than 14 000 clips with the size of 1000 × 1000 pixels and 0.6 overlap ratio to form the training set. In the inference phase, this article uses a sliding window of 1000 × 1000 pixels with an overlap ratio of 0.4. Both in the training and inference phase, this article only feeds the image clips to the neural network. The outputs of all image clips are stitched together to present the final results of the large-scale SAR image.

C. Evaluation Metrics

Followed by Everingham *et al.* [71], there are three broadly accepted criteria for evaluating detection outcomes, average

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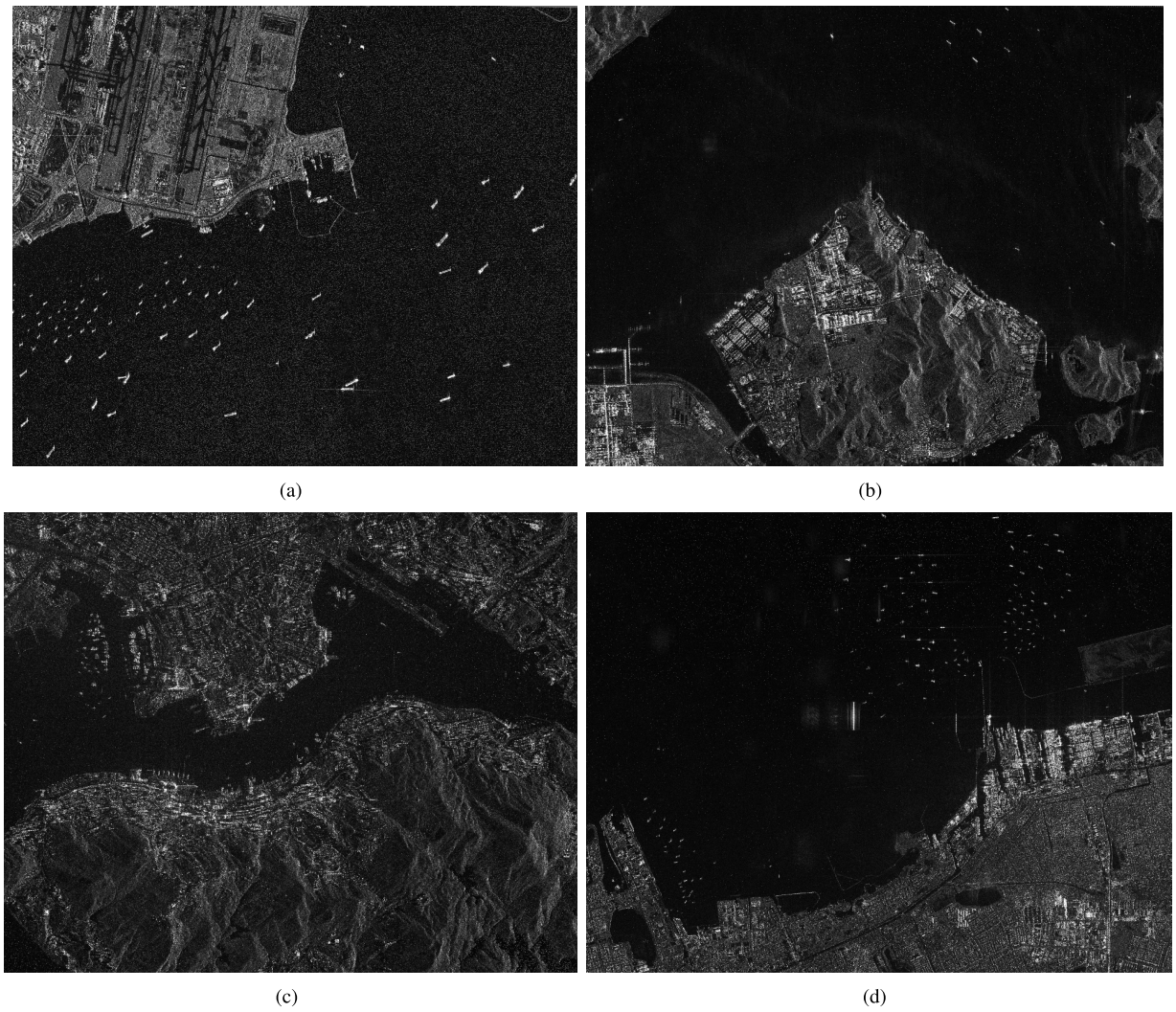

Fig. 6. Image examples in the LDSD dataset. (a)–(d) Numerous SAR scenes in LDSD, such as inshore scene, offshore scene, island scene, land scene, and so on.

precision (AP), precision-recall curve (PRC), and $F1$-score. First, every predicted bounding box B_{p} calculates the IoU with the ground-truth bounding box B_{gt}, and the IoU of two oriented bounding box is defined as follows:

$$\mathrm{IoU} = \frac{\mathrm{area}(B_{\mathrm{p}} \cap B_{\mathrm{gt}})}{\mathrm{area}(B_{\mathrm{p}} \cup B_{\mathrm{gt}})}. \tag{13}$$

Additionally, precision and recall are still adopted to evaluate the detection results, and they are computed as follows:

$$\mathrm{precision} = \frac{N_{\mathrm{TP}}}{N_{\mathrm{TP}} + N_{\mathrm{FP}}} \tag{14}$$

$$\mathrm{recall} = \frac{N_{\mathrm{TP}}}{N_{\mathrm{TP}} + N_{\mathrm{FN}}} \tag{15}$$

where N_{TP}, N_{FP}, and N_{FN} denote the number of true positives (TP), false positives (FP), and false negatives (FN), respectively. TP corresponds to the correctly detected ships whose IoU with B_{gt} are higher than 0.5. FP represents the false alarms or the predicted boxes whose IoU with B_{gt} are lower than 0.5. FN is the missing ship. All outcomes are sorted by decreasing order of the detection score. PRC can be drawn after calculating the precision and recall in the sorted results. Additionally, the AP metric is the most frequent metric to assess the performance of a detector. AP metric is calculated according to the area under the PRC for each category. Obviously, in this article, ship is the only class of detector. Thus, this article only calculates the AP of the ship to assess the ability of our method. According to [71], the AP metric is defined as the integral under the corresponding PRC as

$$\mathrm{AP} = \int \mathrm{Precision}|_{\mathrm{recall}=r}. \tag{16}$$

If AP is calculated at the IoU threshold of 0.5, it can be defined as $\mathrm{AP}_{0.5}$. Similarly, $\mathrm{AP}_{0.75}$ denotes the AP calculated at the IoU threshold of 0.75, which needs higher localization accuracy to further evaluate the detection ability of the method.

Precision and recall cannot fully reflect the performance of the detector. Therefore, $F1$ is introduced to combine the precision and recall metrics to a single measure in order to evaluate the performance of a detector, which is defined as follows:

$$\mathrm{F1} = \frac{2 \times \mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}. \tag{17}$$

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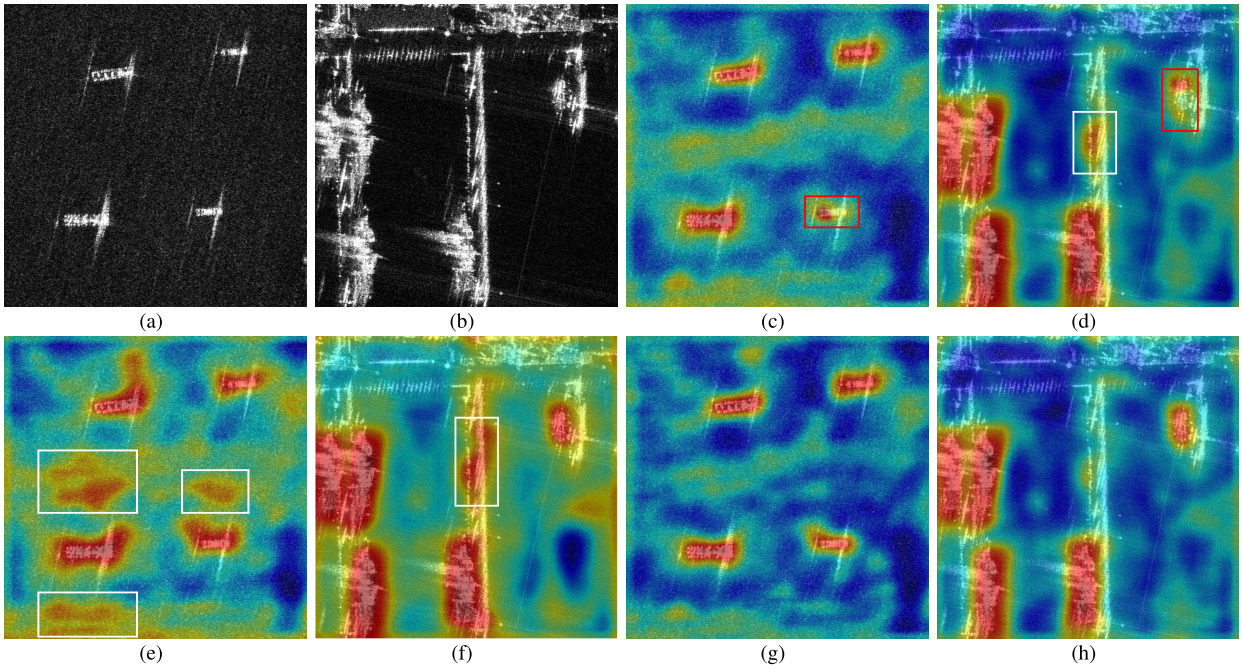

Fig. 7. Visualization of the confidence heatmaps. (a) and (b) Origin SAR images. (c) and (d) Confidence heatmaps without ACS module. (e) and (f) Confidence heatmaps without the ship attention module. (g) and (h) Confidence heatmaps with ship attention and ACS module. The red and white rectangles correspond to the partially detected ships and false alarms, respectively.

D. Ablation Experiments

To evaluate the performance of the proposed method and analyze the influence of each designed component, this article has carried out a series of ablation experiments. In the ablation experiment, the ACS and ship attention module are removed from the proposed method. The results of each trial are put together for comparison. Without specific notes, all the experiment settings are the same.

1) Effect of Each Module: We first investigate the impact of the ACS module by adding it to the baseline. The performance of the baseline with the ACS is shown in Table II. The ACS module has a gain of 0.02 on $AP_{0.5}$, 0.017 on $AP_{0.75}$, 0.0134 on precision, and 0.0203 on recall. The ablation experimental results in the second row of Table II show that the ACS module achieves comparable performance. The inshore ships exist in a complex background and have multiple sizes. Therefore, this module cascades different receptive field atrous convolutions to extract multi-scale semantic information features around ship strong scattering points. Benefitting from extracting the semantic information of multi-scale ships and the surrounding background, the ACS module enables the network to increase the location accuracy of the ship feature points and achieve higher promotion on recall compared with the precision index.

We investigate the impact of the ship attention module by adding it to the baseline. In the third row of Table II, the ship attention module has a gain of 0.034 on $AP_{0.5}$, 0.021 on $AP_{0.75}$, 0.0301 on precision, and 0.0091 on recall. As shown in the last row of Table II, with the help of both ACS module and ship attention module, the performance has a gain of 0.043 on $AP_{0.5}$, 0.041 on $AP_{0.75}$, 0.0473 on precision, and 0.0381 on recall. The inshore ships are surrounded by water and wharves, and the land area looks similar to the ships. As the ship attention module can suppress the image feature texture with high repetition, this module can cluster the scattering points belonging to the ships and suppress those on land, which may result in false alarms. Benefitting from the ship attention module, the proposed method achieves higher promotion on precision compared with recall. Additionally, the ACS module does not need supervision information and can automatically learn the texture features of strong scattering points distribution between ship and land. The above advantages enhance the robustness of the network and reduce the interference of land.

TABLE II
INFLUENCE OF EACH COMPONENT IN THE PROPOSED METHOD

ACS	Ship attention	Precision	Recall	F1	$AP_{0.5}$	$AP_{0.75}$
×	×	0.9011	0.9023	0.9016	0.8874	0.6421
✓	×	0.9145	0.9226	0.9185	0.9075	0.6592
×	✓	0.9312	0.9114	0.9211	0.9215	0.6632
✓	✓	**0.9484**	**0.9404**	**0.9443**	**0.9304**	**0.6831**

2) Analysis of Each Module on the Heatmaps: Fig. 7 shows some visualized results to demonstrate an intuitive comprehension of the effect of two submodules. The generated heatmaps are the outputs from the detection module. In order to reflect the detection performance of the proposed method for multi-size ships, Fig. 7 displays the heatmap results of multi-size ships. As displayed in Fig. 7(c) and (d), without the ACS model, the proposed method cannot get adequate semantic features and long-range context information for small-size ships, which leads to the partial detection result of small-size ships. The small-size ship here refers to the relatively small ship compared with the ship in Fig. 7(b).

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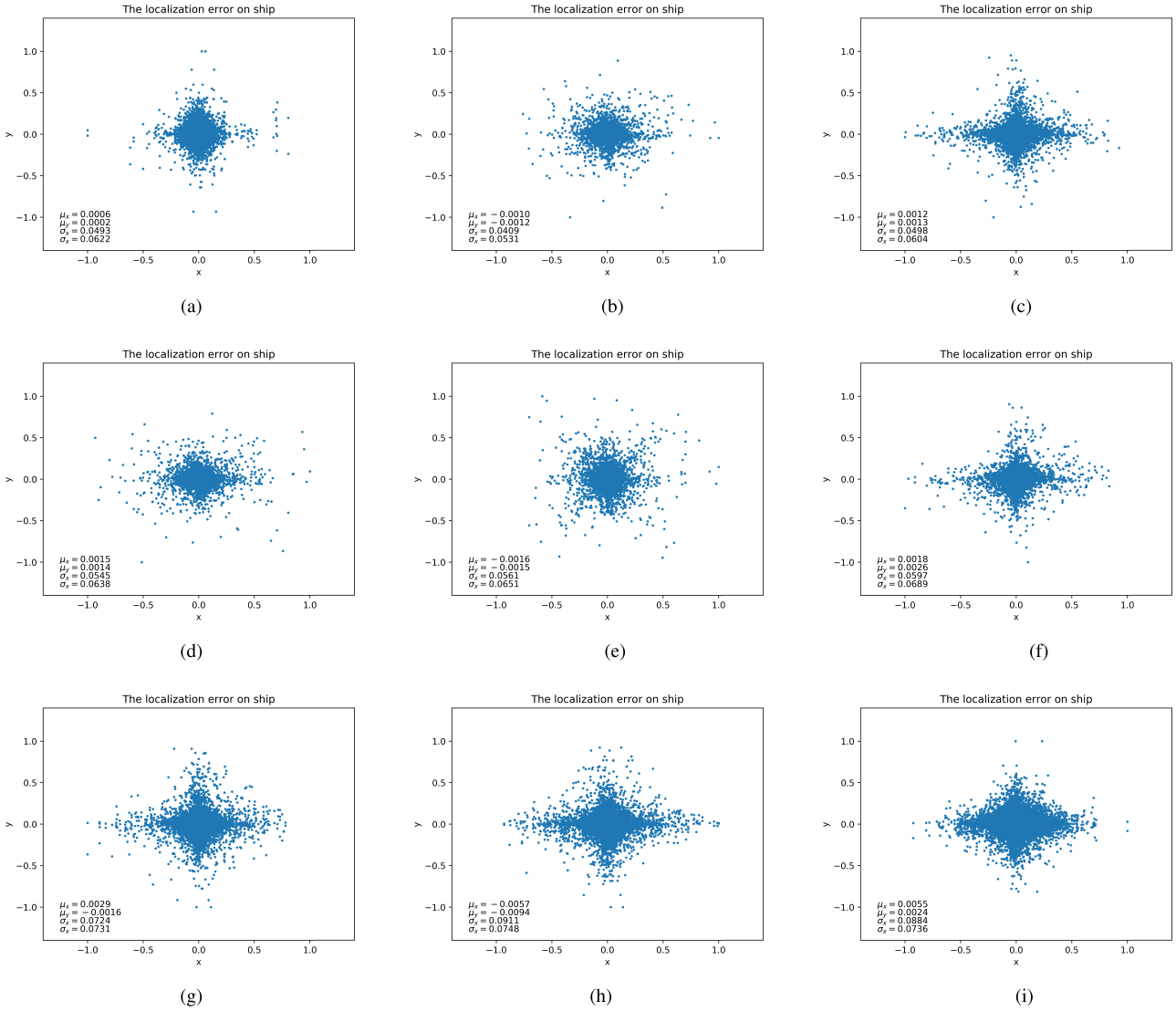

Fig. 8. Analysis of localization errors between nine different methods. The variables μ_x, μ_y, δ_x, and δ_y in the legend represent the averages of localization error and the standard deviations of localization error of different axes. (a) Our method. (b) CR2A-Net. (c) Quad-FPN. (d) SDGH-Net. (e) RetinaNet. (f) FCOS. (g) Mask-RCNN. (h) Faster-RCNN. (i) SSD(300).

As displayed in Fig. 7(e) and (f), without the ship attention model, the proposed method learns rough texture features of strong scattering points, which leads to inshore false alarm under the strong interference of harbor facilities. Due to the two modules mentioned above, the network focuses more on multi-scale ships and eliminates the interference of harbor facilities to a certain degree. Therefore, the robustness of the proposed method and the detection capability of multi-scale ships are promoted.

E. Performance Analysis of Oriented Object Detectors

1) Hyperparameter Analysis: In (3), the hyperparameter α and β are initialized as 0. In the training process, the proposed method gradually assigns value [65] to α and β. At first, the learnable α and β allow the network to rely on the information in the local neighborhood. Then the network gradually learns to assign more weight to the non-local evidence. This helps the network learn the simple task first and then progressively increase the complexity of the task. The loss function in (9) sums the pull loss, push loss, and localization loss. Here, the hyperparameters ϵ_1 and ϵ_2 control the balance between the pull loss, push loss, and localization loss. We consider that the contributions of pull loss and push loss are equal. Therefore, both ϵ_1 and ϵ_2 are set to 0.1. To investigate the training loss, we conduct experiments on the proposed method by varying the hyperparameter combinations. Table III reports the evaluation results on the LDSD. $AP_{0.5}$ in Table III is calculated based on the oriented results. The model is insensitive to the values of ϵ_1 and ϵ_2 in a wide range.

2) Comparison of Localization Error: To explore the localization precision of the methods mentioned above, the localization error between the prediction and ground truth is calculated over large-scale SAR images, which is defined as the distance between the four points of the predicted box and the four

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TABLE III

PERFORMANCE OF DETECTOR UNDER DIFFERENT HYPERPARAMETER COMBINATIONS

(ϵ_1, ϵ_2)	(0.01, 0.01)	(0.1, 0.1)	(1, 1)	(10, 10)	(100, 100)
$AP_{0.5}$	0.921	0.930	0.926	0.913	0.904

TABLE IV

COMPARISON OF DIFFERENT CNN-BASED METHODS (ORIENTED BOUNDING BOX RESULT)

Method	Anchor-free	Stage	Precision	Recall	F1	$AP_{0.5}$	$AP_{0.75}$
SSD(300)	×	One	0.3081	0.5192	0.3871	0.2454	0.1013
RetinaNet	×	One	0.5721	0.5471	0.5605	0.3134	0.1153
FCOS	✓	One	0.5542	0.5323	0.5430	0.3024	0.1092
Faster R-CNN	×	Two	0.3913	0.5244	0.4471	0.2685	0.1031
Mask R-CNN	×	Two	0.5411	0.5233	0.5322	0.2994	0.1193
SDGH-Net	✓	One	0.5357	0.5816	0.5536	0.3258	0.1159
Quad-FPN	✓	Two	0.5728	0.6014	0.5868	0.3567	0.1258
CR2A-Net	×	Two	0.9151	0.9203	0.9176	0.9197	0.6625
Proposed Method	✓	One	**0.9484**	**0.9404**	**0.9443**	**0.930**	**0.683**

points of the ground truth box. It is worth mentioning that the value is standardized by the size of the object. The comparison result of localization precision is shown in Fig. 8. The metrics include the mean of localization error (μ_x and μ_y) along the horizontal axis and vertical axis, respectively. And the standard deviations of localization error (δ_x and δ_y) along the same axis. The standard deviations of localization error reflect the localization ability of the detector.

The localization precision of the proposed method is compared with other CNN-based methods (designed for object detection in the natural scene, optical remote sensing, and SAR images), including CR2A-Net [72], Quad-FPN [73], SDGH-Net [74], Faster-RCNN, Mask-RCNN, FCOS, RetinaNet, and SSD. Faster-RCNN and Mask-RCNN are two-stage detectors, while RetinaNet, FCOS, and SSD are one-stage detectors. The SDGH-Net and CR2A-Net are horizontal and oriented ship detectors in optical remote sensing, respectively. The Quad-FPN is an innovative CNN-based method, which is specially designed for SAR ship detection. In Fig. 8, SSD and Faster-RCNN suffer from poor detection capability and have a larger localization error than our method. FCOS, RetinaNet, and Mask-RCNN have better performance, but the detection ability is still insufficient due to the lack of SAR characteristics. Compared with SDGH-Net, Quad-FPN, and CR2A-Net, the proposed method achieves more competitive performance and has the smallest standard deviations of the localization error. Therefore, we can conclude that the proposed method has a stronger detection capability on oriented ship detection.

3) Comparison of Horizontal and Oriented Results: This article uses both the oriented boxes and the horizontal boxes (converted from the oriented boxes) to train the above detectors and the proposed method. In the predicted stage, this article calculates the AP with the oriented and horizontal results, respectively. It is noteworthy that when evaluating oriented AP, horizontal detectors use the horizontal boxes as they cannot generate the oriented boxes. The oriented experimental detection results are displayed in Table IV, and the corresponding results of horizontal boxes are demonstrated in Table V. It is

TABLE V

COMPARISON OF DIFFERENT CNN-BASED METHODS (HORIZONTAL BOUNDING BOX RESULT)

Method	Anchor-free	Stage	Precision	Recall	F1	$AP_{0.5}$	$AP_{0.75}$
SSD(300)	×	One	0.5322	0.8501	0.6543	0.6946	0.1918
RetinaNet	×	One	0.9011	0.8961	0.8984	0.8867	0.6898
FCOS	✓	One	0.8852	0.8731	0.8782	0.8605	0.6617
Faster R-CNN	×	Two	0.6611	0.8583	0.7473	0.7916	0.2989
Mask R-CNN	×	Two	0.7773	0.8922	0.8312	0.8725	0.4365
SDGH-Net	✓	One	0.8775	0.8816	0.8795	0.8523	0.6564
Quad-FPN	✓	Two	0.8849	0.8895	0.8872	0.8732	0.6721
CR2A-Net	×	Two	0.9167	0.9044	0.9106	0.9113	0.6966
Proposed Method	✓	One	**0.9421**	**0.9341**	**0.9372**	**0.9294**	**0.7118**

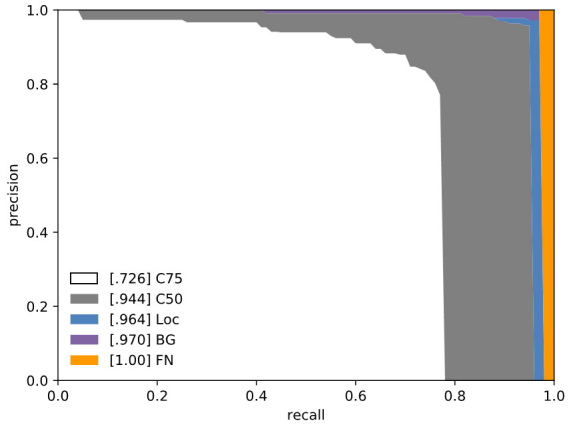

Fig. 9. Detailed analysis of detector performance for our method in oriented results.

shown that our method increases the precision and recall rate in both oriented and horizontal results. $AP_{0.75}$ is a strict metric in object detection defined by the COCO dataset [75], which reflects the localization capability of the proposed method. According to $AP_{0.75}$ in Fig. 9, the proposed method represents an advantage of accurate localization ability. Because our method combines the SAR characteristics and regresses the ship's oriented boxes by detecting strong scattering points. Moreover, when it comes to the horizontal bounding box in Table V, the performance superiority is still obvious in both precision and recall rate. In order to further verify the generalization ability of the proposed method, all the detectors are experimented on SSDD. Table VI illustrates that our method still achieves a superior performance. Based on the comparison in both LDSD and SSDD results, our approach outperforms the above-mentioned detectors and achieves state-of-the-art performance, in which the effectiveness of strong scattering points network is verified.

4) Analysis of PRC: Based on the analysis method of object detection in [75] and [76], this article utilizes several curves, C75, C50, localization error curve (Loc), and confusion with background curve (BG) to further analysis the strong scattering points network, which is shown in Fig. 9. C75 denotes PRC at IoU = 0.75 (AP at strict IoU) and the area under the curve corresponds to $AP_{0.75}$ metric. C50 denotes PRC at IoU = 0.5 and the area under the curve corresponds to $AP_{0.75}$ metric. Loc denotes PRC at IoU = 0.1 where the localization error results are ignored, except for duplicate detections. Localization error is caused when an object is detected by a misaligned box

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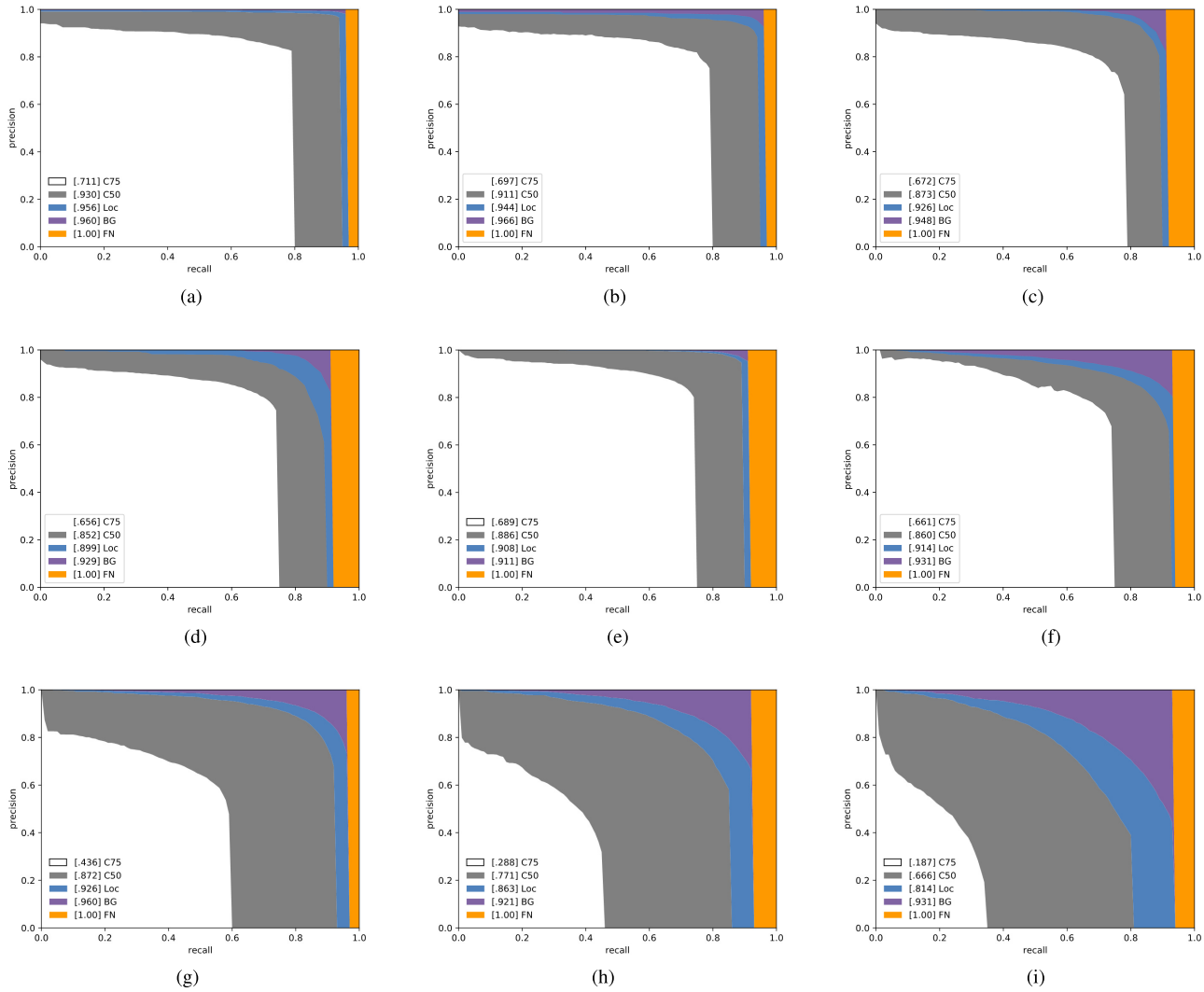

Fig. 10. Detailed PRC for all methods in horizontal results. The legends of different colors in each figure represent the integral under the corresponding curve. (a) Our method. (b) CR2A-Net. (c) Quad-FPN. (d) SDGH-Net. (e) RetinaNet. (f) FCOS. (g) Mask-RCNN. (h) Faster-RCNN. (i) SSD(300).

TABLE VI
COMPARISON OF DIFFERENT CNN-BASED METHODS ON SSDD (HORIZONTAL BOUNDING BOX RESULT)

Method	Precision	Recall	F1	$AP_{0.5}$	$AP_{0.75}$
SSD(300)	0.7078	0.7989	0.7506	0.7658	0.3526
RetinaNet	0.8915	0.9025	0.8969	0.9136	0.5937
FCOS	0.8901	0.8923	0.8912	0.8995	0.5722
Faster R-CNN	0.8513	0.8158	0.8331	0.8039	0.4719
Mask R-CNN	0.8627	0.8821	0.8722	0.8875	0.5642
SDGH-Net	0.8515	0.8967	0.8736	0.8733	0.5712
Quad-FPN	0.8896	0.9397	0.9139	0.9325	0.6957
CR2A-Net	0.9388	0.9401	0.9394	0.9365	0.7127
Proposed Method	**0.9489**	**0.9404**	**0.9446**	**0.9377**	**0.7239**

which has a trivial overlap (IoU $\in [0.1, 0.5)$) with the ground truth. The Loc is drawn to measure the impact of localization error on the detectors. The blue area in each plot reflects the performance degradation of the detector caused by localization error. For the FP examples (false alarm), the box has an overlap of less than 0.1 with any ground-truth box and it is labeled as confusion with the background. After removing all the false alarm examples, the new PRC called BG is drawn in the figure. FN denotes the PRC after all remaining errors are removed (trivially AP $= 1$). The white area denotes the AP calculated by the bounding boxes with IoU > 0.5. And the gray area denotes the AP calculated by the bounding boxes with IoU $\in [0.5, 0.75]$. The bule area denotes the AP improvement based on the assumption that the detection boxes with IoU < 0.5 are replaced by the ground truth. Then, the purple area denotes the AP increment by removing false alarms. At last, the orange area denotes the AP promotion by adding missing detection.

The area under each curve is shown in brackets in the legend. Fig. 9 displays that the overall AP at IoU $= 0.75$ is 0.726, and AP at IoU $= 0.5$ is 0.944. From the above results, we can draw a conclusion that our network has performance superiority in SAR ship detection. Fig. 10 exhibits the analysis PRC of all the methods in horizontal results. The results

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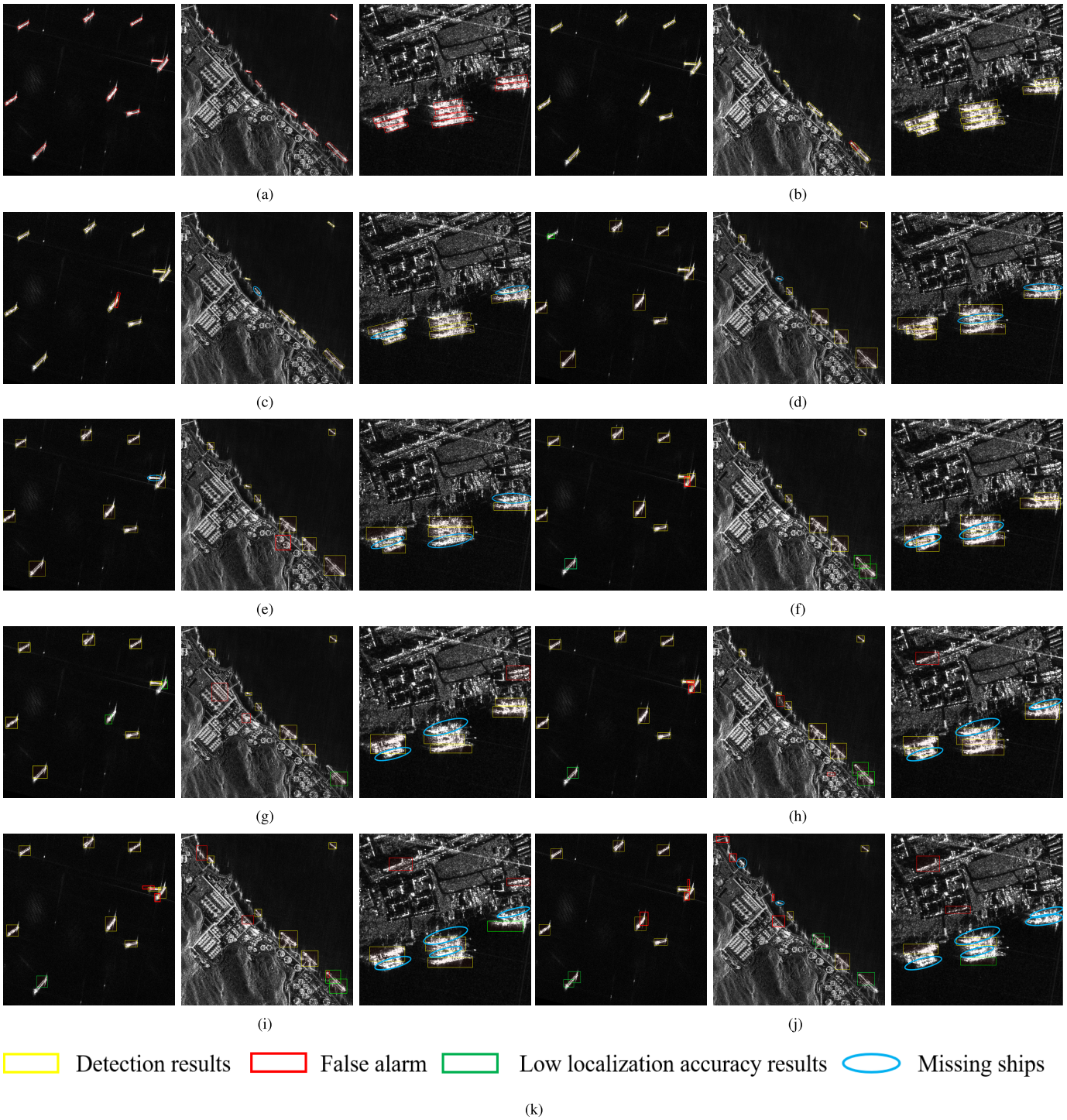

Fig. 11. All method detection results on the LDSD dataset. The yellow rectangles, green rectangles, red rectangles, and blue circles correspond to detection results, low localization accuracy results, false alarms, and missing ships, respectively. (a) Ground truth. (b) Proposed Method. (c) CR2A-Net. (d) Quad-FPN. (e) SDGH-Net. (f) RetinaNet. (g) FCOS. (h) Mask-RCNN. (i) Faster-RCNN. (j) SSD. (k) Legend for each color box.

demonstrate that even if the oriented results are converted into horizontal results, the method still has the best capability in localization, false alarm rate, and lowest missing detection rate. At the same time, the effects of Loc, BG, and FN are as follows. First, the localization error (Loc) has a certain effect on the detection results. Corresponding to the purple area, after replacing all the low localization boxes (IoU $<$ 0.5) to the ground truth, and the AP is increased by 0.02. Second, BG has a slight effect on ship detection. As displayed in the blue area, after removing the false alarms, AP has a small increase of 0.006. Third, the detector suffers most from FN among the three factors. The orange area indicates that the

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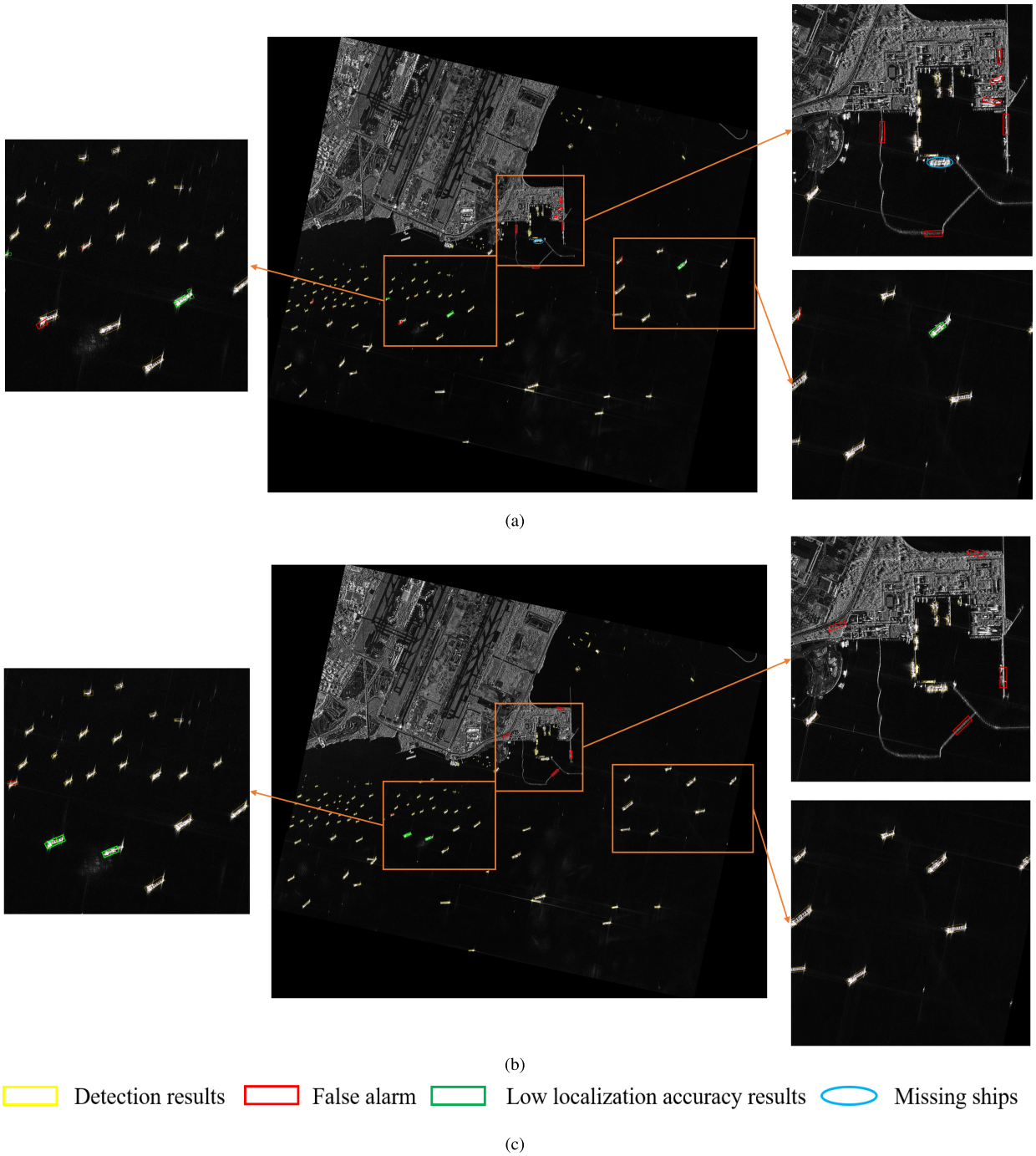

Fig. 12. Detection results in large-scale SAR images. (a) Results of CR2A-Net. (b) Results of our method. (c) Legend for each color box. The yellow rectangles, green rectangles, red rectangles, and blue circle are the detection results, low localization accuracy, false alarms, and missing ships, respectively.

missing detection result in the AP decrease by 0.03, which leads to the decrease of AP higher than the sum of BG and Loc effects.

F. Visualization Analysis of Oriented Object Detector

1) Comparison of Visualization Results: The detection results of all the mentioned methods are visualized in Fig. 11. We follow the previous work [51] which analyzes the performance of the algorithm in densely docked ships scene on SAR-Ship-Dataset [77]. Fig. 11 displays three typical ship detection application scenes, including open sea, inshore, and densely docked ships. Faster R-CNN and SSD miss some inshore densely docked ships and generate some false alarms on land. Meanwhile, RetinaNet and Mask-RCNN achieve higher recall rates, but there are some false alarms on land and some low localization accuracy boxes still exist. The Quad-FPN misses an inshore small-scale ship and SDGH-Net mistakenly detects the inshore artificial facility as ship. The oriented detector CR2A-Net still misses an inshore ship and generates a false alarm object. At the same time, they also miss some inshore densely docked ships. In contrast, our network obtains a better performance and higher localization

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accuracy to some extent. Meanwhile, the proposed method has a remarkable ability to detect the densely docked ships. As shown in Fig. 11(e) and (g)–(j), the red rectangles on land are the inshore artificial facilities with highly similar texture with ships, which results in false alarms. Due to the ship attention module, the proposed method has fewer false alarms on land and verifies its anti-interference ability.

2) Detection Result in Large-Scale SAR Image: As CR2A-Net has the second-best performance in the above comparison, the results of CR2A-Net and the proposed method are visualized in order to analyze the detection performance in the large-scale SAR image. These two algorithms are tested on an SAR image of Changyi port, Singapore, which are displayed in Fig. 12. First, due to the combination of ship characteristics in the SAR image and the ship attention module, the proposed method has fewer false alarms in the inshore area than CR2A-Net. Second, benefitting from the ACS module, our method has fewer missing detection of multi-scale ships in the inshore scene than CR2A-Net. On the whole, our method achieves better performance and represents stronger anti-interference capability than the CR2A-Net detector in the large-scale SAR image.

Though the strong scattering points network has a satisfactory detection performance, there still exist several limitations. The anti-interference ability needs to be further improved, as there are still several false alarms on land in Fig. 12. In addition, some convex wharves are difficult to distinguish from ships. Both the false alarm and missing detection problems will be investigated in our future work. There are still other sources of strong scattering points in the SAR image which need to be further eliminated, such as ghosts of land, cross-sidelobes of ships, islands, and oil platforms. Eliminating these interferences to get a more accurate oriented box is another focus of our future work.

V. Conclusion

In this article, a novel approach for oriented ship detection based on the SAR imaging mechanism and target characteristics is proposed. First, our method extracts the strong scattering points in the SAR images and aggregates them to form the ship's arbitrary orientation box, which is an anchor-free end-to-end detector. Second, the ACS module is designed to effectively capture the multi-scale information of ships to improve the location accuracy. Third, the ship attention module can be used to suppress the interference of strong scattering points on land. Compared with the commonly used CNN-based methods, the strong scattering points network shows the performance superiority in ship detection precision rate, false alarm rate, and location accuracy. Finally, a series of experiments on the LDSD and SSDD demonstrate the effectiveness of the proposed method.

References

[1] L. Du, H. Dai, Y. Wang, W. Xie, and Z. Wang, "Target discrimination based on weakly supervised learning for high-resolution SAR images in complex scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 461–472, Jan. 2020.

[2] K. Fu, F.-Z. Dou, H.-C. Li, W.-H. Diao, X. Sun, and G.-L. Xu, "Aircraft recognition in SAR images based on scattering structure feature and template matching," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 11, pp. 4206–4217, Nov. 2018.

[3] B. Li, B. Liu, W. Guo, Z. Zhang, and W. Yu, "Ship size extraction for Sentinel-1 images based on dual-polarization fusion and nonlinear regression: Push error under one pixel," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4887–4905, Aug. 2018.

[4] L. Du, L. Li, D. Wei, and J. Mao, "Saliency-guided single shot multibox detector for target detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3366–3376, May 2020.

[5] L. Huang *et al.*, "OpenSARShip: A dataset dedicated to Sentinel-1 ship interpretation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 1, pp. 195–208, Jan. 2018.

[6] A. Kostinski and W.-M. Boerner, "On foundations of radar polarimetry," *IEEE Trans. Antennas Propag.*, vol. AP-34, no. 12, pp. 1395–1404, Dec. 1986.

[7] T. Hamasaki, L. Ferro-Famil, E. Pottier, and M. Sato, "Applications of polarimetric interferometric ground-based SAR (GB-SAR) system to environment monitoring and disaster prevention," in *Proc. Eur. Radar Conf. (EURAD)*, 2005, pp. 29–32.

[8] W.-M. Boerner, "Recent advances in extra-wide-band polarimetry, interferometry and polarimetric interferometry in synthetic aperture remote sensing and its applications," *IEEE Proc.-Radar Sonar Navigat.*, vol. 150, no. 3, pp. 113–124, Jun. 2003.

[9] Z. Deng, H. Sun, S. Zhou, and J. Zhao, "Learning deep ship detector in SAR images from scratch," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 4021–4039, Jun. 2019.

[10] J. Zhao, W. Guo, Z. Zhang, and W. Yu, "A coupled convolutional neural network for small and densely clustered ship detection in SAR images," *Sci. China Inf. Sci.*, vol. 62, no. 4, p. 42301, 2019.

[11] Z. Cui, Q. Li, Z. Cao, and N. Liu, "Dense attention pyramid networks for multi-scale ship detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8983–8997, Nov. 2019.

[12] W. Liu *et al.*, "SSD: Single shot MultiBox detector," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 21–37.

[13] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[14] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 2, pp. 318–327, Feb. 2020.

[15] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[16] K. He, G. Gkioxari, P. Dollár, and R. B. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[17] H. Dai, L. Du, Y. Wang, and Z. Wang, "A modified CFAR algorithm based on object proposals for ship target detection in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 12, pp. 1925–1929, Dec. 2016.

[18] X. Wang and C. Chen, "Ship detection for complex background SAR images based on a multiscale variance weighted image entropy method," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 2, pp. 184–187, Feb. 2017.

[19] S. Wang, M. Wang, S. Yang, and L. Jiao, "New hierarchical saliency filtering for fast ship detection in high-resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 1, pp. 351–362, Jan. 2017.

[20] G. Gao and G. Shi, "CFAR ship detection in nonhomogeneous sea clutter using polarimetric SAR data based on the notch filter," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4811–4824, Aug. 2017.

[21] M. Kang, K. Ji, X. Leng, and Z. Lin, "Contextual region-based convolutional neural network with multilayer fusion for SAR ship detection," *Remote Sens.*, vol. 9, no. 8, pp. 860–872, 2017.

[22] J. Jiao *et al.*, "A densely connected end-to-end neural network for multiscale and multiscene SAR ship detection," *IEEE Access*, vol. 6, pp. 20881–20892, 2018.

[23] C. Chen, C. He, C. Hu, H. Pei, and L. Jiao, "A deep neural network based on an attention mechanism for SAR ship detection in multiscale and complex scenarios," *IEEE Access*, vol. 7, pp. 104848–104863, 2019.

[24] Y. Zhao, L. Zhao, C. Li, and G. Kuang, "Pyramid attention dilated network for aircraft detection in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 4, pp. 662–666, Apr. 2021.

[25] J. Li, C. Qu, and J. Shao, "Ship detection in SAR images based on an improved faster R-CNN," in *Proc. SAR Big Data Era, Models, Methods Appl. (BIGSARDATA)*, Nov. 2017, pp. 1–6.

[26] L. M. Novak, M. C. Burl, and W. W. Irving, "Optimal polarimetric processing for enhanced target detection," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 29, no. 1, pp. 234–244, Jan. 1993.

[27] D. Stagliano, A. Lupidi, and F. Berizzi, "Ship detection from SAR images based on CFAR and wavelet transform," in *Proc. Tyrrhenian Workshop Adv. Radar Remote Sens. (TyWRRS)*, Sep. 2012, pp. 53–58.

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[28] J. He, Y. Wang, H. Liu, N. Wang, and J. Wang, "A novel automatic PolSAR ship detection method based on superpixel-level local information measurement," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 3, pp. 384–388, Mar. 2018.

[29] K. Eldhuset, "An automatic ship and ship wake detection system for spaceborne SAR images in coastal regions," *IEEE Trans. Geosci. Remote Sens.*, vol. 34, no. 4, pp. 1010–1019, Jul. 1996.

[30] C. C. Wackerman, K. S. Friedman, W. G. Pichel, P. Clemente-Colón, and X. Li, "Automatic detection of ships in RADARSAT-1 SAR imagery," *Can. J. Remote Sens.*, vol. 27, no. 5, pp. 568–577, 2001.

[31] J. N. Kapur, P. K. Sahoo, and A. K. C. Wong, "A new method for gray-level picture thresholding using the entropy of the histogram," *Comput. Vis., Graph., Image Process.*, vol. 29, no. 3, pp. 273–285, 1985.

[32] M. D. Henschel, M. T. Rey, J. W. M. Campbell, and D. Petrovic, "Comparison of probability statistics for automated ship detection in SAR imagery," in *Proc. 3rd Int. Conf. Appl. Photonic Technol., Closing Gap Theory, Develop., Appl.*, Dec. 1998, pp. 986–991.

[33] C.-H. Lin, Y.-C. Lin, Y. Bai, W.-H. Chung, T.-S. Lee, and H. Huttunen, "DL-CFAR: A novel CFAR target detection method based on deep learning," in *Proc. IEEE 90th Veh. Technol. Conf. (VTC-Fall)*, Sep. 2019, pp. 1–6.

[34] J. He, Y. Wang, and H. Liu, "Ship classification in medium-resolution SAR images via densely connected triplet CNNs integrating Fisher discrimination regularized metric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3022–3039, Apr. 2021.

[35] H. Law and J. Deng, "CornerNet: Detecting objects as paired keypoints," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 734–750.

[36] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9626–9635.

[37] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.

[38] J. Dai, Y. Li, K. He, and J. Sun, "R-FCN: Object detection via region-based fully convolutional networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2016, pp. 379–387.

[39] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.

[40] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.

[41] Y. Li, Y. Chen, N. Wang, and Z.-X. Zhang, "Scale-aware trident networks for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6054–6063.

[42] J. Pang, K. Chen, J. Shi, H. Feng, W. Ouyang, and D. Lin, "Libra R-CNN: Towards balanced learning for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 821–830.

[43] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.

[44] J. Ding, N. Xue, Y. Long, G.-S. Xia, and Q. Lu, "Learning RoI transformer for oriented object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2844–2853.

[45] P. Wang, X. Sun, W. Diao, and K. Fu, "FMSSD: Feature-merged single-shot detection for multiscale objects in large-scale remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3377–3390, Dec. 2020.

[46] X. Sun, P. Wang, C. Wang, Y. Liu, and K. Fu, "PBNet: Part-based convolutional neural network for complex composite object detection in remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 173, pp. 50–65, Mar. 2021.

[47] X. Sun, Y. Liu, Z. Yan, P. Wang, W. Diao, and K. Fu, "SRAF-Net: Shape robust anchor-free network for garbage dumps in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6154–6168, Jul. 2021.

[48] X. Sun, A. Shi, H. Huang, and H. Mayer, "BAS^{4}Net: Boundary-aware semi-supervised semantic segmentation network for very high resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5398–5413, 2020.

[49] K. Fu, Z. Chang, Y. Zhang, and X. Sun, "Point-based estimator for arbitrary-oriented object detection in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4370–4387, May 2021.

[50] J. Fu *et al.*, "An anchor-free method based on feature balancing and refinement network for multiscale ship detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1331–1344, Feb. 2021.

[51] Z. Cui, X. Wang, N. Liu, Z. Cao, and J. Yang, "Ship detection in large-scale SAR images via spatial shuffle-group enhance attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 379–391, Jan. 2021.

[52] Y. LeCun *et al.*, "Backpropagation applied to handwritten zip code recognition," *Neural Comput.*, vol. 1, no. 4, pp. 541–551, 1989.

[53] P. Sermanet, D. Eigen, X. Zhang, M. Mathieu, R. Fergus, and Y. Lecun, "OverFeat: Integrated recognition, localization and detection using convolutional networks," in *Proc. Int. Conf. Learn. Represent.*, Dec. 2013, pp. 1–16.

[54] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[55] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[56] M. D. Zeiler, G. W. Taylor, and R. Fergus, "Adaptive deconvolutional networks for mid and high level feature learning," in *Proc. Int. Conf. Comput. Vis.*, Nov. 2011, pp. 2018–2025.

[57] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.

[58] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder-decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.

[59] K. Grauman and T. Darrell, "The pyramid match kernel: Discriminative classification with sets of image features," in *Proc. ICCV*, vol. 2. Oct. 2005, pp. 1458–1465.

[60] K. He, X. Zhang, S. Ren, and J. Sun, "Spatial pyramid pooling in deep convolutional networks for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 9, pp. 1904–1916, Sep. 2015.

[61] C. Szegedy *et al.*, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1–9.

[62] M. Lin, Q. Chen, and S. Yan, "Network in network," in *Proc. Int. Conf. Learn. Represent.*, 2014, pp. 1–10.

[63] C. Peng, X. Zhang, G. Yu, G. Luo, and J. Sun, "Large kernel matters—Improve semantic segmentation by global convolutional network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1743–1751.

[64] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6230–6239.

[65] H. Zhang, I. Goodfellow, D. Metaxas, and A. Odena, "Self-attention generative adversarial networks," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 7354–7363.

[66] E. Jakeman and P. N. Pusey, "A model for non-Rayleigh sea echo," *IEEE Trans. Antennas Propag.*, vol. AP-24, no. 6, pp. 806–814, Nov. 1976.

[67] Q. Jiang *et al.*, "Ship detection in RADARSAT SAR imagery," in *Proc. IEEE Int. Conf. Syst., Man, Cybern.*, vol. 5, Oct. 1998, pp. 4562–4566.

[68] J. Jao, "Amplitude distribution of composite terrain radar clutter and the K-distribution," *IEEE Trans. Antennas Propag.*, vol. AP-32, no. 10, pp. 1049–1062, Oct. 1984.

[69] A. Newell, Z. Huang, and J. Deng, "Associative embedding: End-to-end learning for joint detection and grouping," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 2277–2287.

[70] A. Neubeck and L. Van Gool, "Efficient non-maximum suppression," in *Proc. Int. Conf. Pattern Recognit.*, vol. 3, Aug. 2006, pp. 850–855.

[71] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The Pascal visual object classes (VOC) challenge," *Int. J. Comput. Vis.*, vol. 88, no. 2, pp. 303–338, Sep. 2010.

[72] Y. Yu, X. Yang, J. Li, and X. Gao, "A cascade rotated anchor-aided detector for ship detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, early access, Dec. 7, 2021, doi: 10.1109/TGRS.2020.3040273.

[73] T. Zhang, X. Zhang, and X. Ke, "Quad-FPN: A novel quad feature pyramid network for SAR ship detection," *Remote Sens.*, vol. 13, no. 14, p. 2771, Jul. 2021.

[74] Z. Wang, Y. Zhou, F. Wang, S. Wang, and Z. Xu, "SDGH-Net: Ship detection in optical remote sensing images based on Gaussian heatmap regression," *Remote Sens.*, vol. 13, no. 3, p. 499, Jan. 2021.

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[75] T.-Y. Lin *et al.*, "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 740–755.
[76] D. Hoiem, Y. Chodpathumwan, and Q. Dai, "Diagnosing error in object detectors," in *Proc. Eur. Conf. Comput. Vis.*, 2012, pp. 340–353.
[77] Y. Wang, C. Wang, H. Zhang, Y. Dong, and S. Wei, "A SAR dataset of ship detection for deep learning under complex backgrounds," *Remote Sens.*, vol. 11, no. 7, p. 765, Mar. 2019.

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